

■ Integrate

`Integrate[f, x]` gives the indefinite integral $\int f dx$.

`Integrate[f, {x, xmin, xmax}]` gives the definite integral $\int_{xmin}^{xmax} f dx$.

`Integrate[f, {x, xmin, xmax}, {y, ymin, ymax}]` gives the multiple integral $\int_{xmin}^{xmax} dx \int_{ymin}^{ymax} dy f$.

Multiple integrals use a variant of the standard iterator notation. The first variable given corresponds to the outermost integral, and is done last. ■ `Integrate` can evaluate integrals of rational functions. It can also evaluate integrals that involve exponential, logarithmic, trigonometric and inverse trigonometric functions, so long as the result comes out in terms of the same set of functions. ■ `Integrate` carries out some simplifications on integrals it can explicitly do. ■ You can get a numerical result by applying `N` to a definite integral. ■ You can assign values to patterns involving `Integrate` to give results for new classes of integrals. ■ The integration variable can be any expression. However, `Integrate` uses only its literal form. The object $d(x^n)$, for example, is not converted to $nx^{n-1}dx$. ■ See page 419. ■ See also: `NIntegrate`.